Does Proximity Feedback Modality Reduce Falls in a Non-Dominant Hand Tilt Task?

PathSense: A Preliminary Study

Eugene Sy • June 2, 2026

One Hand Busy, One Hand Blind

Your brain can't control what it can't sense.
Elderly shopping · Surgical precision · Drone piloting: same principle.

Everyday Cane + shopping bag

Surgical Robotic arm precision

Precision Drone under stress

Guiard (1987): non-dominant hand is the **frame-setter**—it localizes position but *not* fine motion. Without feedback, control is **impossible**.

Why Feedback Channel Matters

No Feedback		**With Feedback**
(Can't learn)	$\longrightarrow$	(Brain adapts)
BLIND		✓ AWARE

Body Schema

Maravita & Iriki (2004): Brain extends into tools held in hand

Haptic Advantage

Cuppone et al. (2016): Touch improves proprioceptive acuity

Audio Trade-off

Sigrist et al. (2013): Sound may overload attention without clear mapping

Which channel helps non-dominant hand control?

Research Questions

RQ1* Does modality affect fall counts?

RQ2 Does feedback improve performance (baseline-adjusted)?

RQ3 Which modality shows better improvement?

RQ4 Does difficulty interact with feedback?

The Question: Does modality choice predict performance?
The Finding: Baseline and difficulty matter more.

*Primary. RQ2–3 address the critical discovery: baseline control differs 10× by modality.

PathSense: Study Design

Participant + display + controller

Cohort Demographics

N	25 participants
Age	17.3 ± 5.2 yrs (13–47)
Gender	48% M, 48% W
Handedness	92% right-handed
Modality	9 per condition

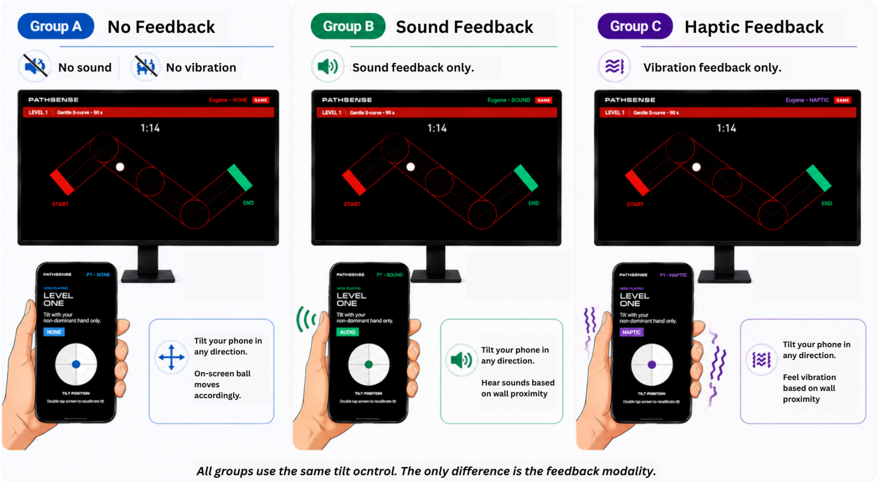
Difficulty Levels

L1 Practice | L2 Easy | L3 Medium | L4 Hard

Measurement

• Falls • Proximity • 10 Hz gyro

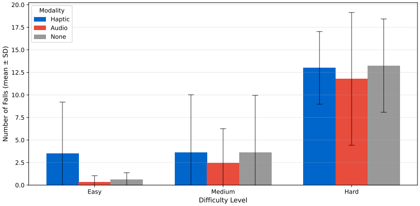
The Three Feedback Conditions



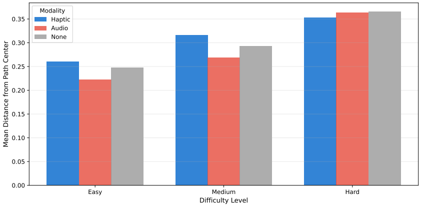
All groups: same tilt control, same game, same difficulty levels. Only difference: feedback modality.

Results: Control Accuracy & Fall Rate

Fall Counts



Path Deviation



RQ1 Statistics:

- Kruskal-Wallis: $p > 0.19$ at all levels (L2–L4)
- Effect sizes (rank-biserial r): −0.44 to +0.25 (negligible)
- Mann-Whitney pairwise (Holm corrected): all $p > 0.27$
- Bayesian: $\mathrm{BF}_{01} \approx 84$ (Easy), 7 (Medium), 5.4 (Hard)
- Sample sizes: $n = 8$–9 per modality per level

Results: Baseline Confound & Proximity Response

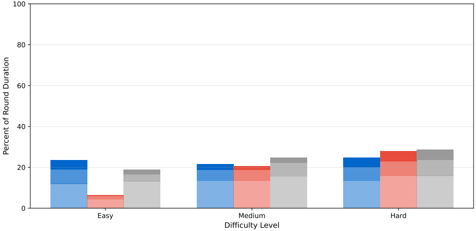
Practice Baseline (Before Feedback)

Group	Falls
Audio	0.33 ✓
None	2.14
Haptic	4.88 ×

10× gap

Groups unequal before testing

Proximity Response (Stacked)



Modalities: Haptic | Audio | None

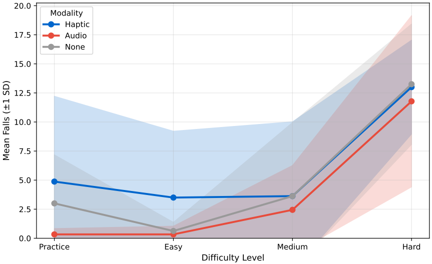
Zones: Light (safe) | Medium (warn) | Dark (danger)

Baseline Test (Practice): Mann-Whitney $U = 28$, $p = 0.04$, $r = -0.57$ (moderate effect)
RQ2 (Change-from-baseline): Haptic Δ = −1.38, Audio Δ = 0.00, None Δ = −2.38

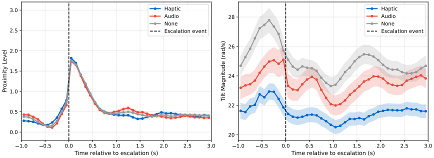
RQ3: Haptic shows relative improvement; audio flat. Small absolute differences.

Results: Learning Trajectory & Steering Response

Learning Curve



Time-Locked Correction



RQ4 (Difficulty × Modality): All groups: improve at Easy (Δ ≈ −1 to −2 falls), deteriorate at Hard (Δ ≈ +8 to +11 falls). Kruskal-Wallis interaction test: $p > 0.19$.

Steering Response: No modality advantage in time-to-recovery or correction magnitude. Difficulty is the dominant factor.

What We Actually Found

Finding 1: Baseline Varies Dramatically

10× gap in baseline control before any feedback. Groups weren't equal from start.

Finding 2: Modality Doesn't Predict Outcomes

No modality effect at group level. Haptic ≠ Audio ≠ None (all $p > 0.19$).

Bayesian analysis: $\mathrm{BF}_{01} \approx 84$ (strong evidence for null).

Finding 3: Task Difficulty Dominates

All groups improve at Easy, crash at Hard. Difficulty is the real limiter.

> *"It's a good training for elders or students with special needs."*
> — Participant feedback (on audio condition)

Next step: Individual-level analysis to understand responsiveness patterns.

What We Learned

RQ1:	No modality effect on absolute falls
RQ2/RQ3:	Baseline responsiveness predicts improvement
RQ4:	Task difficulty, not feedback, dominates

This study said "No" to modality.
But that "No" led to a better question:
Who IS responsive to which feedback?

Next: Scale sample. Test more populations. Continue building.
Insight: Measure baseline first. It predicts everything.

Refs: Guiard (1987) • Maravita & Iriki (2004) • Cuppone et al. (2016)